

RL for decision making in unstructured environments, applications in congested traffic and laneless environments

1st Ahmad Assaad 2nd Prof. Naseem Daher

Department of Mechanical Engineering

American University of Beirut

Beirut, Lebanon

aha173@mail.aub.edu, nd38@mail.aub.edu

Abstract—Reinforcement learning has become widely used in autonomous decision-making problems, where agents must constantly adapt to changing environments. One such problem is autonomous ground vehicle navigation, particularly in environments where the structure of normal highway driving begins to break down. In this paper, we focus specifically on two such scenarios: congested traffic environments and environments with no lane markings. The objective is to study deep reinforcement learning as the high-level decision-making component and evaluate how well it performs under these two forms of reduced environmental structure. The innovation in this work lies in extending existing structured highway simulators toward these unstructured driving scenarios and formulating both problems within a reinforcement learning decision-making framework.

Index Terms—Reinforcement Learning, Autonomous Ground Vehicles, Decision-Making, Congested Traffic, Laneless Driving, Highway Simulation.

I. INTRODUCTION

There has been significant development in the performance of autonomous vehicles when environments are structured and driving behavior is predictable. However, an important question remains: what happens when this structure breaks down and driving behavior becomes irregular? In developing countries, autonomous vehicles still face major challenges due to unstructured driving conditions, including congestion, weak or absent lane markings, mixed vehicle types, poor road conditions, and unpredictable driver behavior [1]. These conditions motivate the need for adaptive decision-making methods. Classical methods such as hard-coded logic, fuzzy logic, and model predictive control can become difficult to scale or tune under irregular traffic, while learning-based methods introduce their own challenges in computation, problem formulation, reward design, safety, and generalization.

Reinforcement learning is promising because it allows agents to learn decision-making policies directly from interaction with the environment. However, most reinforcement learning formulations for autonomous driving still assume structured environments and lane-based traffic. These assumptions become stressed in two important cases: congestion and absence of lane markings. Congestion increases the number of relevant surrounding vehicles and stresses the observation

space, reward design, and interaction reasoning of the agent. In contrast, the absence of lane markings breaks the lane-based structure used by standard traffic models, requiring a reformulation of both the environment and the decision-making problem.

These two scenarios are chosen because they represent two progressively harder forms of reduced structure. Congestion preserves the lane-based road layout, but increases interaction complexity by introducing more relevant surrounding vehicles, rear-flow pressure, and closer vehicle spacing. It therefore tests whether existing reinforcement learning formulations can remain effective when the environment becomes dense but still structurally familiar. In contrast, the absence of lane markings removes the discrete lane structure assumed by most highway simulators. In this case, the problem is no longer simply lane selection, but continuous corridor selection, requiring a reformulation of the traffic model, driver behavior, and reinforcement learning problem.

In this paper, we focus on using deep reinforcement learning for high-level decision-making under these two unstructured scenarios: congested traffic and no-lane-marking environments. The motivation behind studying these two cases is to first evaluate how reinforcement learning agents fail in increasingly dense but still lane-based environments, and then study how the problem must be reformulated when the lane structure itself is removed. This raises a broader question about generalization across different forms of reduced structure and whether a single decision-making architecture can remain effective across such scenarios.

The contributions of this paper are twofold. First, we study congested highway traffic as a partially unstructured setting, analyze failure modes of baseline reinforcement learning agents, and propose extensions to the observation and reward formulation to improve decision-making under dense interaction. Second, we formulate a laneless highway environment by shifting from lane-based driver behavior to corridor-based interaction, and then define the corresponding reinforcement learning problem in this laneless setting.

II. RELATED WORKS

A. Main approaches in autonomous decision-making literature

Autonomous driving decision-making has traditionally been studied through planning, prediction, and control-based methods. Zhang et al. survey motion planning for self-driving vehicles in intersection scenarios, showing how classical planning frameworks reason about structured road geometry and traffic rules [2]. Hubmann et al. address decision-making under interaction and uncertain prediction of surrounding vehicles, emphasizing the need to account for other agents' future behavior [3]. Model predictive control has also been widely used for autonomous vehicle motion control because it allows vehicle motion to be optimized while enforcing physical and safety constraints [4]. Cooperative and game-theoretic MPC methods further extend this idea to interactive scenarios such as intersections and highway merging [5], [6]. However, these approaches generally rely on explicit models, predefined constraints, and sufficiently structured environments, making them harder to adapt when traffic becomes irregular or road structure is weakly defined.

A second direction uses reinforcement learning as the main decision-making method. Liu et al. propose a heuristics-oriented reinforcement learning approach for overtaking, where reward design is used to guide safety and efficiency [7]. Sakib studies highway lane changing under uncertainty using deep reinforcement learning [8], while Alizadeh et al. use DRL for automated lane-change decision-making in dynamic highway traffic [9]. PPO-based formulations have also been used for automated lane changing, extending RL decision-making toward policy-gradient methods and continuous control ideas [10]. Other highway tasks, such as highway exiting and hierarchical self-driving decision-making, have also been formulated using reinforcement learning [11], [12]. These works show that RL can learn useful driving strategies without manually specifying every decision rule, but most still assume lane-based roads, structured action spaces, and observation representations built around predefined traffic positions.

A third major direction combines reinforcement learning with traditional planning or control in an upper/lower-level architecture. Hoel et al. combine planning and deep reinforcement learning for tactical decision-making, using learning to support high-level decisions while retaining structured planning components [13]. Safe RL with MPC has been studied for autonomous lane merging, where reinforcement learning handles decision-making while MPC improves safety and feasibility [14]. Similar DRL-MPC combinations have been proposed for coordinated ramp merging and interactive autonomous navigation [15], [16]. Robust MPC has also been used as a safety-oriented layer for reinforcement learning in autonomous vehicle motion planning [17]. These hybrid methods address some safety limitations of standalone RL, but they still commonly preserve structured assumptions such as lanes, road geometry, and predictable interaction models.

This leads to the gap addressed in this work: existing approaches improve autonomous driving decision-making in structured or semi-structured environments, but most remain grounded in lane-based traffic assumptions. This leaves open the question of how reinforcement learning behaves when highway structure is progressively weakened through congestion and then removed more fundamentally through the absence of lane markings.

Existing reinforcement learning formulations can also be viewed as attempts to teach the ego vehicle specific driving skills. Overtaking, lane changing, highway exiting, merging, and intersection negotiation are commonly treated as separate decision-making problems, each with its own state representation, action space, and reward design [7]–[12]. These works are useful because they show how reinforcement learning can be shaped toward particular driving behaviors, either through discrete tactical actions or continuous control. However, this skill-based structure also reveals a limitation: the learned policy is often tied to the assumptions of the task formulation, such as lane availability, fixed surrounding-vehicle positions, or a specific traffic scenario. This motivates the need to study whether reinforcement learning decision-making can remain effective when the environment becomes less structured.

In this work, we build on this literature by studying two formulations of reduced structure. In congested environments, the road remains lane-based, but the decision problem becomes harder due to dense interaction, rear-flow pressure, and the larger number of relevant vehicles. For this case, we test reinforcement learning formulations with modified observations, reward shaping, and an adaptive lower-level velocity controller. In laneless environments, the lane structure itself is removed, so the problem must be reformulated from discrete lane selection to continuous corridor selection.

B. Simulation Frameworks for Autonomous Driving Decision-Making

Several simulation frameworks have been developed to evaluate autonomous driving decision-making methods. Highway-env provides a lightweight and open-source environment for testing reinforcement learning agents in structured driving scenarios such as highways, roundabouts, intersections, and parking [21]. Its main advantage is simplicity: it exposes standard observation and action formulations, supports common RL benchmarks, and allows rapid modification of traffic density, vehicle behavior, rewards, and road structure. For this reason, highway-env is used throughout this work as the base environment for validating ideas and extending structured highway scenarios toward congested and laneless settings.

BARK is another behavior and motion planning benchmark designed for multi-agent driving scenarios [36]. Its strength lies in its modularity, since it separates behavior models, scenario generation, planning algorithms, and evaluation tools. This makes it useful for comparing different behavior-planning methods under controlled scenarios. However, its formulation still relies heavily on structured road representations and pre-

TABLE I: Main Research Directions in Autonomous Driving Decision-Making Literature

Research Direction	Literature	Role in This Work
Classical planning, prediction, and model-based control	[2]–[6], [18], [19]	Provides conservative decision-making and control methods, but often depends on structured road geometry, explicit models, and predefined constraints.
Standalone reinforcement learning for autonomous driving skills	[7]–[12], [20]–[23]	Learns specific driving behaviors such as overtaking, lane changing, exiting, and highway navigation, but usually assumes lane-based traffic structure.
Hybrid reinforcement learning and control-based safety architectures	[13]–[17], [24]–[26]	Combines learned decision-making with traditional control layers such as MPC or PID to improve feasibility and safety.
Interaction-aware, dense-traffic, and uncertainty-based decision-making	[27]–[32]	Addresses multi-agent interaction, intention prediction, attention, and uncertainty, motivating the study of congestion.
Simulation frameworks, traffic models, and unstructured-environment studies	[1], [33]–[37]	Provides the environment and traffic-modeling basis for this work, while exposing the limitation of lane-based simulator assumptions.

defined scenario logic, which limits its direct use for studying environments where lane structure becomes weak or absent.

SMARTS is a scalable multi-agent reinforcement learning platform designed for training and evaluating autonomous driving agents in interactive traffic [35]. Its strength is large-scale multi-agent simulation, where multiple agents can be trained or evaluated together under diverse traffic conditions. This makes it useful for studying interaction, emergent behavior, and multi-agent learning. However, like other major simulators, it still largely assumes structured road networks and lane-based traffic organization.

Overall, these simulators provide strong foundations for autonomous driving research, but most are designed around structured roads, lane-based interactions, and predefined traffic models. Since the goal of this work is to study decision-making under reduced structure, highway-env is selected as the base platform because it is simple, open-source, and easily extendable. This allows us first to stress-test existing lane-based formulations under congestion, and then to modify the environment itself toward a laneless highway formulation.

III. BACKGROUND INFORMATION

A. Reinforcement Learning Formulation

Reinforcement learning formulates decision-making as an interaction between an agent and an environment. At each time step t , the agent observes a state s_t , selects an action a_t , receives a reward r_t , and transitions to a new state s_{t+1} . This interaction is commonly modeled as a Markov Decision Process (MDP), defined by the tuple

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma), \quad (1)$$

where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $P(s'|s, a)$ is the transition probability, $R(s, a)$ is the reward function, and $\gamma \in [0, 1]$ is the discount factor.

The return from time step t is defined as

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k}. \quad (2)$$

The objective of reinforcement learning is to find a policy $\pi(a|s)$ that maximizes the expected return,

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi}[G_t]. \quad (3)$$

The state-value function is defined as

$$V^{\pi}(s) = \mathbb{E}_{\pi}[G_t \mid s_t = s], \quad (4)$$

while the action-value function is defined as

$$Q^{\pi}(s, a) = \mathbb{E}_{\pi}[G_t \mid s_t = s, a_t = a]. \quad (5)$$

B. Deep Q-Networks

Deep Q-Networks (DQN) extend tabular Q-learning by approximating the action-value function using a neural network. Instead of storing a value for every state-action pair, DQN learns a parameterized function

$$Q(s, a; \theta), \quad (6)$$

where θ represents the neural network parameters.

The DQN update is based on the Bellman optimality target

$$y = r + \gamma \max_{a'} Q(s', a'; \theta^-), \quad (7)$$

where θ^- are the parameters of a target network. The network is trained by minimizing the temporal-difference loss

$$L(\theta) = \mathbb{E} \left[(y - Q(s, a; \theta))^2 \right]. \quad (8)$$

The resulting policy selects the action with the highest estimated value:

$$a^* = \arg \max_a Q(s, a; \theta). \quad (9)$$

DQN is therefore suitable for discrete decision-making problems, such as selecting high-level highway actions including lane changes, acceleration, deceleration, or maintaining the current behavior.

C. Soft Actor-Critic

While DQN is designed for discrete action spaces, laneless driving requires continuous control over acceleration and steering. For this reason, we also consider Soft Actor-Critic (SAC), an off-policy actor-critic algorithm designed for continuous action spaces.

SAC maximizes not only the expected return, but also the entropy of the policy. The objective is

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t (r_t + \alpha \mathcal{H}(\pi(\cdot|s_t))) \right], \quad (10)$$

where α is the temperature parameter and $\mathcal{H}(\pi(\cdot|s_t))$ is the entropy of the policy at state s_t .

Equivalently, SAC encourages the policy to maximize reward while maintaining exploration. The soft Q-function target is commonly written as

$$y = r + \gamma \mathbb{E}_{a' \sim \pi} [Q(s', a') - \alpha \log \pi(a'|s')]. \quad (11)$$

This makes SAC suitable for the laneless formulation, where the action is continuous and may be represented as

$$a_t = \begin{bmatrix} a_{\text{long}} \\ \delta \end{bmatrix}. \quad (12)$$

D. Highway-Env Vehicle Dynamics

The simulation environment used in this work is based on highway-env. The ego vehicle is modeled using a kinematic bicycle model. The state of the vehicle is represented as

$$x = [X \quad Y \quad \psi \quad v]^T, \quad (13)$$

where X and Y are the vehicle position coordinates, ψ is the heading angle, and v is the longitudinal velocity.

The vehicle dynamics are given by

$$\dot{X} = v \cos(\psi + \beta), \quad (14)$$

$$\dot{Y} = v \sin(\psi + \beta), \quad (15)$$

$$\dot{v} = a, \quad (16)$$

$$\dot{\psi} = \frac{v}{L} \sin(\beta), \quad (17)$$

where a is the longitudinal acceleration, L is the vehicle wheelbase, and β is the slip angle approximation

$$\beta = \tan^{-1} \left(\frac{1}{2} \tan \delta \right), \quad (18)$$

with δ representing the steering angle.

In the standard lane-based formulation, high-level actions are mapped into low-level acceleration and steering commands through internal controllers. This allows reinforcement learning agents to operate at the decision-making level while the environment handles low-level vehicle tracking.

E. Traffic Modeling: IDM and MOBIL

Surrounding vehicles in highway-env are commonly modeled using the Intelligent Driver Model (IDM) for longitudinal motion and the MOBIL model for lane-changing behavior [33], [34]. These models are important in this work because they depend on lane-based notions of leading and following vehicles, which later become limiting in the laneless formulation.

The IDM acceleration is defined as

$$a = a_0 \left[1 - \left(\frac{v}{v_0} \right)^\delta - \left(\frac{s^*(v, \Delta v)}{s} \right)^2 \right], \quad (19)$$

where a_0 is the maximum acceleration, v is the current velocity, v_0 is the desired velocity, s is the gap to the leading vehicle, and Δv is the relative velocity. The desired gap is

$$s^*(v, \Delta v) = s_0 + vT + \frac{v\Delta v}{2\sqrt{a_0 b}}, \quad (20)$$

where s_0 is the minimum gap, T is the desired time headway, and b is the comfortable braking deceleration.

Lane-changing behavior is modeled using MOBIL. A lane change is accepted if the incentive criterion is satisfied:

$$\Delta a_{\text{ego}} + p(\Delta a_{\text{new}} + \Delta a_{\text{old}}) > \Delta a_{\text{th}}, \quad (21)$$

subject to the safety condition

$$\tilde{a}_{\text{new follower}} > -b_{\text{safe}}. \quad (22)$$

Here, Δa_{ego} is the ego vehicle's acceleration gain from changing lanes, Δa_{new} and Δa_{old} represent the acceleration changes imposed on the new and old followers, p is the politeness factor, Δa_{th} is the minimum acceleration gain required for a lane change, and b_{safe} is the maximum acceptable braking imposed on another vehicle.

These equations reveal a key limitation of the standard formulation: both IDM and MOBIL require a well-defined lane structure to identify leading vehicles, following vehicles, and target lanes. In the laneless case, these concepts must be reformulated using geometric corridors rather than discrete lanes.

IV. CONGESTED TRAFFIC RESULTS

A. Baseline DQN Formulation in Congested Traffic

We first consider a plain DQN baseline in a congested highway environment. The purpose of this baseline is not to introduce additional safety mechanisms, but to evaluate how a standard lane-based reinforcement learning formulation behaves when the traffic density is increased. This baseline therefore serves as the reference point for later modifications to the observation space, reward function, attention encoder, and adaptive controller.

1) *Observation Space*: The observation follows the kinematic observation structure used in highway-env. At each time step, the agent receives a fixed-size table

$$s_t \in \mathbb{R}^{V \times F}, \quad (23)$$

where V is the maximum number of observed vehicles and F is the number of features per vehicle. In the congested baseline, the observation is

$$s_t \in \mathbb{R}^{12 \times 5}, \quad (24)$$

corresponding to the ego vehicle and up to eleven surrounding vehicles. Each vehicle row is represented by

$$[\textit{presence}, x, y, v_x, v_y], \quad (25)$$

where *presence* indicates whether a vehicle exists in that row, x and y are the relative positions with respect to the ego vehicle, and v_x and v_y are the relative velocity components. The observation is ego-relative, since absolute coordinates are not used. Vehicles behind the ego are included through rear-aware observation, but no time-to-collision feature, driver-aggressiveness feature, or attention-based encoding is added. The resulting table is flattened internally and passed to a multilayer perceptron DQN.

2) *Action Space*: The agent uses the discrete meta-action space provided by highway-env:

$$\mathcal{A} = \left\{ \begin{array}{l} \text{LANE_LEFT,} \\ \text{IDLE,} \\ \text{LANE_RIGHT,} \\ \text{FASTER,} \\ \text{SLOWER} \end{array} \right\}. \quad (26)$$

These actions are high-level commands rather than direct steering or acceleration inputs. The lane-change actions request a predefined maneuver to the left or right lane, while the speed actions modify the reference velocity. The actual acceleration and steering commands are then generated by the environment's low-level controllers. In particular, longitudinal control follows a proportional structure of the form

$$a = K_p(v_r - v), \quad (27)$$

where v_r is the reference velocity and v is the current ego velocity. Thus, even though DQN selects high-level decisions, the physical execution is handled by the simulator's internal controller.

3) *Reward Function*: The baseline uses the native highway-env reward structure with modified weights. The raw reward is composed of collision, lane preference, and speed components:

$$r_{\text{raw}} = -1.0 \cdot r_{\text{collision}} + 0.1 \cdot r_{\text{right-lane}} + 0.4 \cdot r_{\text{speed}}. \quad (28)$$

The collision term is defined as

$$r_{\text{collision}} = \begin{cases} 1, & \text{if the ego vehicle crashes,} \\ 0, & \text{otherwise.} \end{cases} \quad (29)$$

The speed reward is computed from the forward speed using the configured speed range [20, 30] m/s:

$$r_{\text{speed}} = \text{clip} \left(\frac{v - 20}{30 - 20}, 0, 1 \right). \quad (30)$$

The right-lane reward is given by

$$r_{\text{right-lane}} = \frac{i_{\text{lane}}}{N_{\text{lanes}} - 1}, \quad (31)$$

where i_{lane} is the current lane index and N_{lanes} is the total number of lanes. No explicit lane-change penalty is used in this baseline.

The reward is then normalized from the interval $[-1.0, 0.5]$ to $[0, 1]$ and multiplied by the on-road reward indicator. Therefore, off-road states receive zero reward.

4) *Congested Traffic Configuration*: The baseline is trained in a dense three-lane highway environment. The main configuration parameters are shown in Table II.

Surrounding vehicles follow IDM-based behavior with mixed driver aggressiveness profiles. However, this aggressiveness is not directly exposed to the DQN state. Therefore, the baseline agent must infer surrounding behavior only through relative position and velocity information.

TABLE II: Congested Baseline DQN Environment Configuration

Parameter	Value
Number of lanes	3
Number of vehicles	45
Vehicle density	2.0
Episode duration	40 s
Simulation frequency	15 Hz
Policy frequency	1 Hz
Ego spacing	1.0
Observation shape	12×5
Action space	Discrete meta-actions

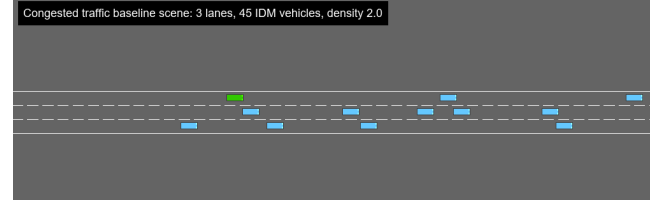


Fig. 1: Snapshot of Congested Traffic Environment

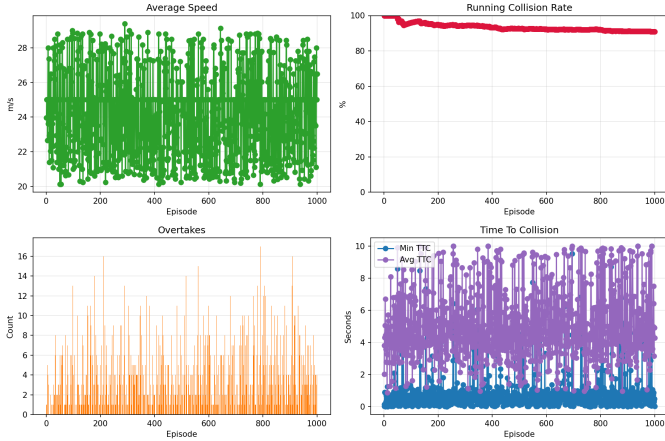
5) *Role of the Baseline*: This baseline intentionally does not include TTC-based safety shaping, attention-based vehicle selection, adaptive velocity control, or explicit driver-type information. It therefore represents the standard lane-based DQN formulation under increased traffic density. Any failure observed in this setting can be attributed to the mismatch between a simple fixed-observation, lane-based RL formulation and the increased interaction complexity of congested traffic.

B. Results

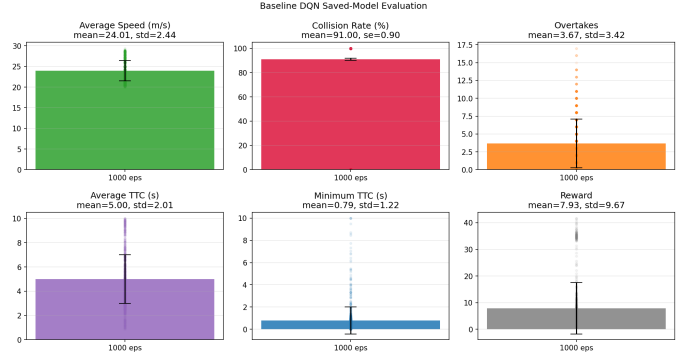
1) *Training and Evaluation Methodology*: For the congested traffic experiments, each DQN configuration was trained across five random seeds for 20,000 timesteps per seed. The best-performing model was selected for detailed saved-model evaluation and collision diagnostics. This selection was used to analyze the strongest policy obtained under each formulation, while recognizing that seed-averaged results would be required for a full statistical comparison.

The desired behavior in congested traffic is not simply to maximize reward, but to maintain the target speed without being overwhelmed by surrounding traffic, avoid unsafe headways, and perform overtaking when necessary. A successful policy is therefore characterized by low collision rate, high average speed, and a reasonable number of overtakes. In addition to reward, evaluation metrics include average speed, collision rate, overtakes, average time-to-collision (TTC), minimum TTC, and collision diagnostics.

The saved model was evaluated deterministically over 1000 episodes using the greedy action selected by the learned Q-function. For a more detailed failure analysis, collision diagnostics were additionally performed over 100 deterministic evaluation episodes. These diagnostics classify whether collisions resulted from poor final action selection, front collision,



(a) Evaluation summary



(b) Evaluation metrics

Fig. 2: Saved-model evaluation of the baseline DQN.

TABLE III: Collision Diagnostics for Baseline DQN in Congested Traffic over 100 Episodes

Diagnostic Category	Episodes	Rate
Bad final action	75	75%
Front collision	14	14%
Side collision	5	5%
No collision	6	6%

TABLE IV: Collision Geometry for Baseline DQN in Congested Traffic

Collision Type	Episodes
Front collision	66
Side collision	27
Rear collision	1
No collision	6

side collision, rear collision, unavoidable rear pressure, or lack of any safe discrete action.

2) *Baseline Failure Analysis*: The baseline DQN performed poorly under congested traffic. In the 100-episode diagnostic evaluation, the agent collided in 94 episodes, corresponding to a collision rate of 94%. The mean episode length before termination was only 13.2 policy steps out of a maximum of 40, indicating that the agent usually failed early in the episode.

The diagnostics suggest that most failures were policy-induced rather than structurally unavoidable. In 75% of diagnostic episodes, the final action selected by the agent was classified as a bad action relative to safer discrete alternatives. Moreover, the rate of *no good discrete action* cases was 0%, and the rate of unavoidable rear-pressure cases was also 0%. This indicates that the agent was usually not trapped in states where every available maneuver was unsafe. Instead, the policy often failed to choose defensive actions early enough.

Most crashes were front-impact collisions, with 66 front collisions compared to 27 side collisions and only 1 rear collision.

This is consistent with insufficient longitudinal decision-making under short headway conditions. The final TTC values further support this interpretation: the mean final front TTC was 1.15 s, while the median final front TTC was only 0.48 s. Therefore, the baseline DQN did not reliably learn to slow down or change lanes before entering critical front-collision situations.

These results show that the standard DQN formulation is not sufficient for congested traffic. Although the observation includes relative vehicle positions and velocities, the agent lacks explicit safety features such as TTC, rear-flow pressure, or interaction-aware vehicle weighting. Similarly, the native reward encourages speed and lane preference but does not strongly penalize unsafe headways before collision. This motivates the later extensions to reward shaping, observation design, and attention-based representations.

C. Extensions to the Congested-Traffic Baseline

The collision diagnostics of the baseline DQN show that most failures are front-impact collisions caused by poor policy decisions rather than unavoidable traffic configurations. This suggests that the standard DQN formulation lacks three key elements: an early safety signal, flexible longitudinal control, and a better way to identify relevant vehicles in dense traffic. We therefore introduce three extensions: a TTC-based reward term, an adaptive TTC-based longitudinal controller, and an ego-attention DQN encoder.

1) *TTC-Based Safety Reward*: The native highway-env reward mainly encourages speed and lane preference, while collision is penalized only after failure occurs. To provide an earlier safety signal, we add a TTC-based shaping term:

$$r_t = r_{\text{base}} + r_{\text{safety}}, \quad (32)$$

where

$$r_{\text{safety}} = b_{\text{TTC}} - p_{\text{low-TTC}} - p_{\text{lag}}. \quad (33)$$

The front TTC is computed from the same-lane gap and closing speed:

$$g_{\text{front}} = s_{\text{front}} - s_{\text{ego}} - \frac{1}{2}(L_{\text{ego}} + L_{\text{front}}), \quad (34)$$

$$TTC_{\text{front}} = \begin{cases} \frac{g_{\text{front}}}{v_{\text{ego}} - v_{\text{front}}}, & v_{\text{ego}} > v_{\text{front}}, \\ TTC_{\text{cap}}, & \text{otherwise.} \end{cases} \quad (35)$$

Low TTC is penalized by

$$p_{\text{low-TTC}} = \min \left(w_{\text{low}} \max \left(0, \frac{T_{\text{safe}} - TTC_{\text{front}}}{T_{\text{safe}}} \right), p_{\text{low,max}} \right), \quad (36)$$

while safe headway is rewarded by

$$b_{\text{TTC}} = \min \left(w_{\text{bonus}} \text{clip} \left(\frac{TTC_{\text{front}} - T_{\text{safe}}}{T_{\text{target}} - T_{\text{safe}}}, 0, 1 \right), b_{\text{max}} \right). \quad (37)$$

To prevent the agent from solving congestion by simply slowing down excessively, a rear-flow lag penalty is added:

$$p_{\text{lag}} = \min (w_{\text{lag}} \Delta v_{\text{deficit}} \mathbb{I}[TTC_{\text{front}} \geq T_{\text{safe}}] \rho_{\text{rear}}, p_{\text{lag,max}}). \quad (38)$$

This reward encourages the vehicle to avoid unsafe front headways while still maintaining traffic flow.

2) *Adaptive TTC-Based Longitudinal Controller*: The baseline meta-actions FASTER and SLOWER map to fixed reference-speed changes, which is too rigid for congested traffic. Instead of moving to a continuous action space, we preserve the discrete DQN action interface and add an adaptive lower-level controller that adjusts the target speed using front TTC.

A smooth TTC score is defined as

$$\sigma_{\text{TTC}} = \frac{1}{1 + \exp \left(-\frac{TTC_{\text{front}} - T_{\text{mid}}}{\tau} \right)}. \quad (39)$$

For FASTER,

$$\Delta v = \Delta v_{\text{fast,max}} \sigma_{\text{TTC}}, \quad (40)$$

and for SLOWER,

$$\Delta v = -[\Delta v_{\text{slow,min}} + (\Delta v_{\text{slow,max}} - \Delta v_{\text{slow,min}})(1 - \sigma_{\text{TTC}})]. \quad (41)$$

The target speed is updated as

$$v_{\text{target,new}} = \text{clip}(v_{\text{target,old}} + \Delta v, v_{\text{min}}, v_{\text{max}}). \quad (42)$$

Thus, when TTC is low, acceleration is suppressed and deceleration is strengthened. When TTC is high, acceleration is allowed. This gives the agent more longitudinal flexibility without adding continuous steering control.

3) *Ego-Attention DQN*: The baseline DQN flattens the kinematic observation, forcing the network to process all vehicles uniformly. In dense traffic, however, only a subset of vehicles is usually relevant to the current decision. To address this, we replace the MLP feature extractor with an ego-attention encoder inspired by social-attention methods [30].

The observation remains

$$O_t \in \mathbb{R}^{N \times F}, \quad (43)$$

with rows

$$[\textit{presence}, x, y, v_x, v_y]. \quad (44)$$

The ego and surrounding vehicles are embedded separately:

$$h_{\text{ego}} = f_{\text{ego}}(o_{\text{ego}}), \quad h_i = f_{\text{other}}(o_i). \quad (45)$$

The ego embedding forms the query, while surrounding vehicles provide keys and values:

$$Q = W_Q h_{\text{ego}}, \quad (46)$$

$$K_i = W_K h_i, \quad (47)$$

$$V_i = W_V h_i. \quad (48)$$

The attention weight for vehicle i is

$$\alpha_i = \text{softmax} \left(\frac{Q K_i^T}{\sqrt{d_h}} \right), \quad (49)$$

and the context vector is

$$c = \sum_i \alpha_i V_i. \quad (50)$$

The final feature vector is

$$z = \frac{1}{2}(W_O c + h_{\text{ego}}), \quad (51)$$

which is passed to the DQN head:

$$Q(s, a) = \text{MLP}(z). \quad (52)$$

This allows the network to learn which surrounding vehicles are most relevant, rather than treating the full traffic table as a flat state vector.

4) *Combined Agent*: The final congested-traffic agent combines the three additions. The TTC reward improves the safety signal, the adaptive controller improves longitudinal flexibility, and the attention encoder improves dense-traffic representation. The DQN remains the high-level decision maker, while the added components make the original discrete formulation better suited for congested traffic.

5) *Ablation Results*: All congested-traffic variants were trained using the same environment configuration, training horizon, evaluation protocol, and DQN hyperparameters as the baseline. The goal of this ablation study is to isolate the effect of each added component: TTC-based reward shaping, ego-attention representation, and adaptive TTC-based longitudinal control. The saved-model evaluation results are summarized in Fig. 3, while the collision diagnostics are shown in Table V.

The baseline DQN achieves high average speed and frequent overtaking, but this behavior is unsafe under congestion. Its

Saved 1k-Episode Evaluation: Behaviour and Safety Metrics

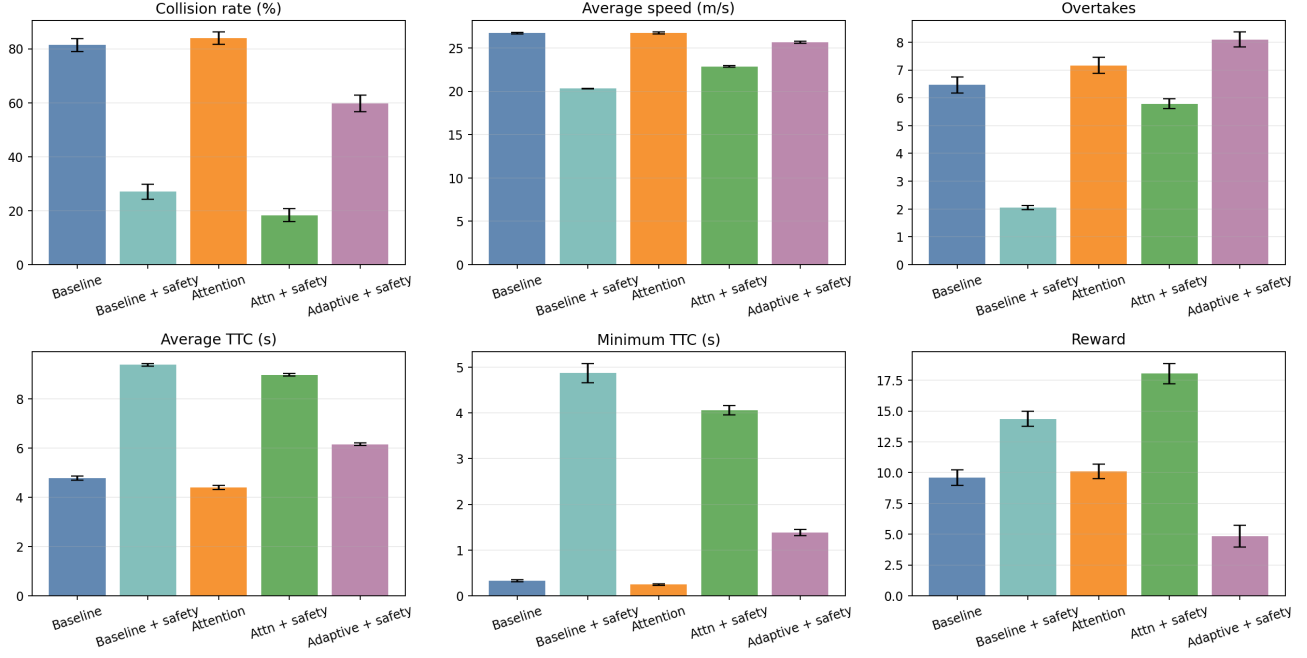


Fig. 3: Saved 1000-episode evaluation metrics for congested-traffic DQN variants.

TABLE V: Collision Diagnostics for Congested-Traffic DQN Variants over 100 Evaluation Episodes

Model	Collisions	Front	Side	Rear	Median Steps to Crash	Median Front TTC at Crash
Baseline DQN v2	99/100	29	69	1	27.0	0.22 s
DQN + TTC Safety Reward v2	22/100	3	19	0	78.5	3.56 s
Attention DQN v2	35/100	10	25	0	83.0	0.27 s
Attention + TTC Safety Reward v2	25/100	2	23	0	78.0	0.57 s
Adaptive + Attention v2	44/100	0	42	2	69.5	0.87 s

collision rate is high and its minimum TTC is very low, showing that the native highway-env reward mainly encourages fast driving without providing enough early safety feedback. The collision diagnostics confirm this failure: the baseline collides in 99/100 diagnostic episodes, with a median front TTC at crash of only 0.22 s.

Adding the TTC safety reward produces the clearest safety improvement. The collision count drops from 99/100 to 22/100, and the median front TTC at crash increases from 0.22 s to 3.56 s. This shows that TTC-based shaping successfully reduces longitudinal front-collision failures. However, this improvement comes with a more conservative driving style, as average speed and overtakes decrease compared to the baseline.

Attention alone improves the agent’s ability to survive longer, but it does not fully solve the safety problem. The attention DQN still produces many collisions, and its median front TTC at crash remains low. This suggests that attention improves representation capacity, but without a safety-aware reward it can still learn aggressive or unsafe decisions.

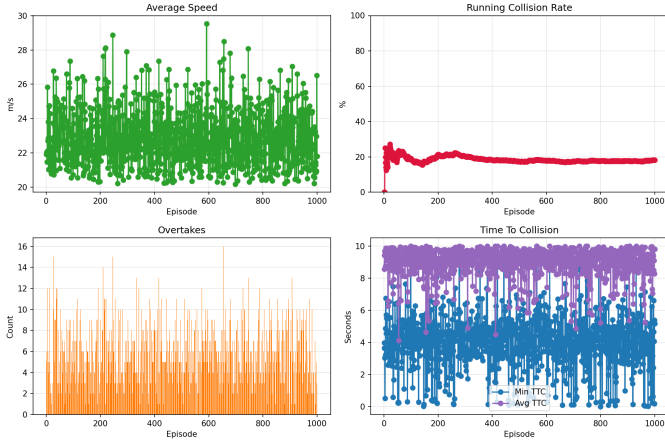
The strongest overall tradeoff appears when attention is combined with TTC safety shaping. This model achieves low collision rate, high average TTC, high minimum TTC, and the

highest reward among the tested variants. The result suggests that attention is most useful when the reward already provides a meaningful safety signal: the reward defines what matters, while attention helps the network identify which vehicles matter.

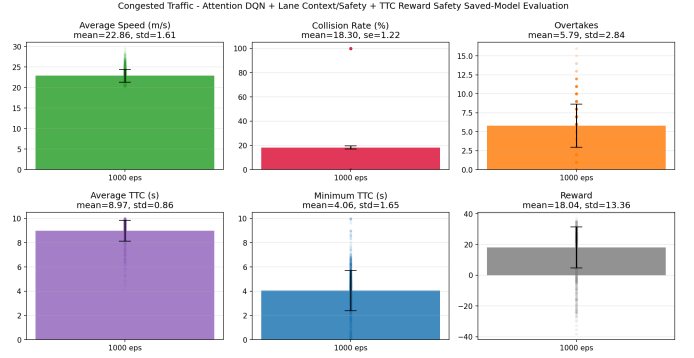
The adaptive TTC-based controller improves longitudinal risk handling, but it does not solve the complete congestion problem. In the collision diagnostics, the adaptive model produces zero front collisions, indicating that TTC-based speed modulation is effective for longitudinal safety. However, it still produces many side collisions, showing that lateral lane-change safety remains unresolved.

Overall, the ablation study shows that the main effect of TTC-based shaping and adaptive speed control is not complete safety, but a transfer of failure mode. The baseline fails through both front and side collisions, while the improved models largely suppress front-TTC failures and leave side collisions as the dominant remaining failure mode. This indicates that congested traffic requires not only longitudinal TTC safety, but also explicit lateral risk modeling, lane-change gap checking, or a safety filter for lane-change actions.

6) *Lane-Context Safety Observation*: The previous ablation showed that TTC-based reward shaping reduced front-collision



(a) Saved-model evaluation of the attention DQN with lane-context safety observation and TTC reward over 1000 episodes.



(b) Episode-level speed, collision rate, overtakes, and TTC trends for the final congested-traffic extension.

Fig. 4: Final congested-traffic extension using attention, TTC reward shaping, and lane-context safety observation.

failures, but the remaining crashes were dominated by side impacts during lane changes. This suggests that the agent required additional information about lateral interaction risk. We therefore extended the observation space with lane-context safety features, including lateral TTC information for neighboring lanes. Unlike the front TTC reward, which only shapes longitudinal following behavior, this observation extension gives the agent direct information about whether adjacent-lane maneuvers are safe.

The resulting agent combines attention, lane-context safety features, and TTC-based reward shaping. As shown in Fig. 4, the model achieves an average speed of 22.86 m/s, a collision rate of 18.30%, an average TTC of 8.97 s, and a minimum TTC of 4.06 s over 1000 deterministic evaluation episodes. The running collision rate stabilizes near 18%, while the speed remains close to the desired highway range. This indicates a better safety-efficiency tradeoff than the previous variants.

7) *Extensions and Future Directions*: The previous ablation showed that TTC-based reward shaping mainly improves longitudinal safety. After adding the safety reward, front collisions decreased significantly, but the remaining failures were dominated by side collisions during lane changes. This indicates that the agent was no longer primarily missing front-headway information, but lateral safety information.

To address this, we tested an additional observation-space extension by adding lane-context safety features, including lateral TTC information for neighboring lanes. This extension directly targets the remaining side-collision failure mode by making adjacent-lane risk observable to the DQN. The resulting model, combining attention, lane-context safety features, and TTC reward shaping, achieved an average speed of 22.86 m/s, collision rate of 18.30%, average TTC of 8.97 s, and minimum TTC of 4.06 s over 1000 deterministic evaluation episodes. This suggests that collision diagnostics can be used not only for evaluation, but also to guide the design of improved RL formulations.

Other extensions were also explored but are left as future work. A SAC-based continuous-control formulation was tested, where the agent directly controlled steering and throttle. However, in the lane-based congested highway setting, this introduced unnecessary control complexity and led to reward-hacking behavior. Since lane changes are already naturally discretized in highway-env, fully continuous steering control was not essential for this part of the problem.

Hierarchical reinforcement learning was also considered as a promising future direction. In principle, an HRL formulation could separate maneuver selection from low-level control: a high-level policy would decide whether to keep lane, overtake, yield, or change lanes, while lower-level policies would control throttle and steering. This separation is especially relevant for congested traffic, where lateral and longitudinal decisions are coupled but may require different forms of control. However, early experiments showed unstable training, so a full HRL formulation is left for future work.

Overall, these extensions suggest that the main path forward is not simply to increase model complexity, but to match the formulation to the observed failure mode. TTC reward addresses longitudinal safety, lane-context observation addresses lateral safety, and future hierarchical architectures may provide a cleaner separation between maneuver choice and low-level control.

V. LANELESS ENVIRONMENT

A. Laneless Highway Environment Formulation

In standard MOBIL, lane changing is formulated over a discrete set of neighboring lanes. For a candidate target lane, the vehicle compares its acceleration before and after the lane change, while also accounting for the effect imposed on surrounding vehicles. The standard MOBIL incentive is

$$G_{\text{MOBIL}} = (\tilde{a}_c - a_c) + p[(\tilde{a}_n - a_n) + (\tilde{a}_o - a_o)], \quad (53)$$

where c is the candidate vehicle, n is the new follower in the target lane, o is the old follower in the current lane, and p is the politeness factor. A lane change is accepted if

$$G_{\text{MOBIL}} \geq \Delta a_{\min}, \quad (54)$$

subject to the safety constraint

$$\tilde{a}_n \geq -b_{\text{safe}}. \quad (55)$$

In the laneless environment, the discrete target lane is replaced by a continuous lateral target coordinate y_k . Instead of evaluating candidate lanes, the vehicle evaluates a finite set of sampled lateral positions,

$$Y = \{y_1, y_2, \dots, y_N\}, \quad (56)$$

with

$$y_k \in \left[-\frac{W}{2} + m, \frac{W}{2} - m \right], \quad (57)$$

where W is the road width and m is a boundary margin.

The main change is that leaders and followers are no longer defined by lane index. For each candidate y_k , surrounding vehicles are selected using corridor overlap. Vehicle j is considered laterally relevant to vehicle i at candidate coordinate y_k if

$$|y_j - y_k| \leq \frac{w_i}{2} + \frac{w_j}{2} + c_{\text{lat}}, \quad (58)$$

where w_i and w_j are vehicle widths and c_{lat} is a lateral clearance buffer. This defines the laneless neighbor function

$$(\text{preceding}(y_k), \text{following}(y_k)). \quad (59)$$

The MOBIL gain is then evaluated for each lateral candidate:

$$\begin{aligned} G(y_k) = & a_c(y_k) - a_c(y_{\text{cur}}) \\ & + p \left[a_n^{\text{after}}(y_k) - a_n^{\text{before}}(y_k) \right. \\ & \left. + a_o^{\text{after}} - a_o^{\text{before}} \right]. \end{aligned} \quad (60)$$

The safety constraint becomes

$$a_n^{\text{after}}(y_k) \geq -b_{\text{safe}}. \quad (61)$$

A lateral swept-region constraint is also added. The swept interval is defined by

$$y_{\min}^{\text{sweep}} = \min(y_{\text{cur}}, y_k) - \frac{w_c}{2} - m, \quad (62)$$

$$y_{\max}^{\text{sweep}} = \max(y_{\text{cur}}, y_k) + \frac{w_c}{2} + m. \quad (63)$$

A candidate is rejected if another vehicle intersects the interval $[y_{\min}^{\text{sweep}}, y_{\max}^{\text{sweep}}]$ within the longitudinal safety window

$$|s_j - s_c| \leq \frac{L_c}{2} + \frac{L_j}{2} + L_{\text{safe}}. \quad (64)$$

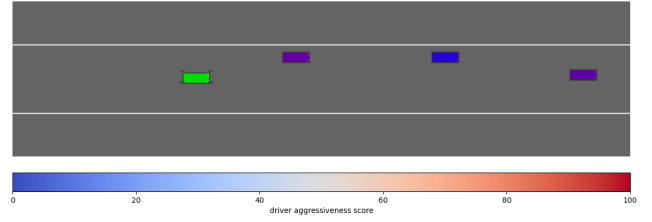


Fig. 5: Snapshot of Laneless Env

The selected laneless target is therefore

$$y_{\text{target}} = \arg \max_{y_k \in Y} G(y_k), \quad (65)$$

subject to

$$G(y_k) \geq \Delta a_{\min}, \quad (66)$$

$$a_n^{\text{after}}(y_k) \geq -b_{\text{safe}}, \quad (67)$$

and no swept-region conflict.

Thus, laneless MOBIL preserves the original MOBIL ideas of acceleration gain, politeness, and safety, but replaces discrete lane-change evaluation with continuous corridor selection:

$$\begin{array}{c} \text{lane change decision} \\ \Downarrow \\ \text{continuous lateral corridor decision} \end{array} \quad (68)$$

B. RL Formulation in the Laneless Environment

The laneless environment changes the decision-making problem fundamentally. In the lane-based highway setting, the ego vehicle chooses from discrete maneuvers such as lane left, lane right, faster, and slower. This is possible because lanes define a finite set of driving corridors. Once lane markings are removed, this structure disappears. The decision is no longer “which lane should I choose?”, but rather “which lateral corridor should I move toward?”

The laneless problem is formulated as an MDP,

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma), \quad (69)$$

where the ego vehicle is controlled by the RL policy, while surrounding vehicles follow the laneless IDM/MOBIL traffic model.

1) *State Space*: The ego vehicle observes a kinematic table,

$$s_t = O_t \in \mathbb{R}^{N \times F}, \quad (70)$$

where each row corresponds to one observed vehicle. In this work, each vehicle is represented by

$$o_i = [p_i, x_i, y_i, v_{x,i}, v_{y,i}], \quad (71)$$

where p_i is the presence flag, (x_i, y_i) are relative positions, and $(v_{x,i}, v_{y,i})$ are relative velocity components.

Compared with the lane-based formulation, the lateral coordinate y_i becomes more important. In the lane-based case, y_i mostly indicates lane position. In the laneless case, y_i defines the continuous corridor geometry around the ego vehicle.

2) *Action Space*: The main change is in the action space. In the lane-based DQN formulation, the action set is discrete:

$$\mathcal{A} = \left\{ \begin{array}{c} \text{LANE_LEFT,} \\ \text{IDLE,} \\ \text{LANE_RIGHT,} \\ \text{FASTER,} \\ \text{SLOWER} \end{array} \right\}. \quad (72)$$

In the laneless formulation, this is replaced by a continuous action:

$$a_t = \begin{bmatrix} u_v \\ u_y \end{bmatrix} \in [-1, 1]^2, \quad (73)$$

where u_v controls the target speed and u_y controls the target lateral position. The environment maps this action into target updates:

$$\Delta v_{\text{target}} = k_v u_v, \quad (74)$$

$$\Delta y_{\text{target}} = k_y u_y. \quad (75)$$

Thus, the ego vehicle does not choose a lane. It continuously updates a desired lateral target:

$$y_{\text{target}} \in \left[-\frac{W}{2} + m, \frac{W}{2} - m \right], \quad (76)$$

where W is the road width and m is a boundary margin.

The low-level controller then tracks the target speed and target lateral position. Therefore, SAC does not directly output steering and acceleration; it outputs continuous commands in the laneless decision space.

3) *Reward Function*: The reward is designed to balance speed, safety, road-boundary behavior, and overtaking:

$$r_t = r_{\text{speed}} + r_{\text{safety}} + r_{\text{spatial}} + r_{\text{overtake}}. \quad (77)$$

The speed term encourages the ego vehicle to maintain highway speed. The safety term penalizes unsafe front interactions using TTC with vehicles that overlap the ego's lateral corridor. The spatial term discourages driving too close to the road boundary or making unnecessary lateral movements. The overtaking term rewards passing slower vehicles when the maneuver is safe.

The important difference from the lane-based reward is that safety is no longer defined only by lane membership. A vehicle is relevant if it overlaps the ego's continuous corridor:

$$|y_j - y_{\text{ego}}| \leq \frac{w_{\text{ego}}}{2} + \frac{w_j}{2} + c_{\text{lat}}. \quad (78)$$

This same idea is used to compute TTC and interaction risk in the laneless setting.

4) *SAC Objective*: Because the action space is continuous, SAC is used instead of DQN. SAC learns a stochastic policy $\pi(a|s)$ that maximizes entropy-regularized return:

$$J(\pi) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t (r_t - \alpha \log \pi(a_t|s_t)) \right]. \quad (79)$$

The entropy term encourages exploration, which is important because the ego must learn both longitudinal and lateral decisions in a continuous corridor.

5) *Interpretation*: The shift from lane-based to laneless driving changes the RL problem in three ways. First, the action space changes from discrete lane maneuvers to continuous lateral-target control. Second, safety is no longer evaluated using lane membership, but through geometric corridor overlap. Third, the environment transition dynamics change because surrounding vehicles also move according to laneless IDM/MOBIL rather than lane-based IDM/MOBIL.

Therefore, the laneless formulation is not simply a denser version of the highway problem. It is a different decision-making problem where the agent must learn continuous corridor selection instead of discrete lane selection.

C. Laneless SAC Results

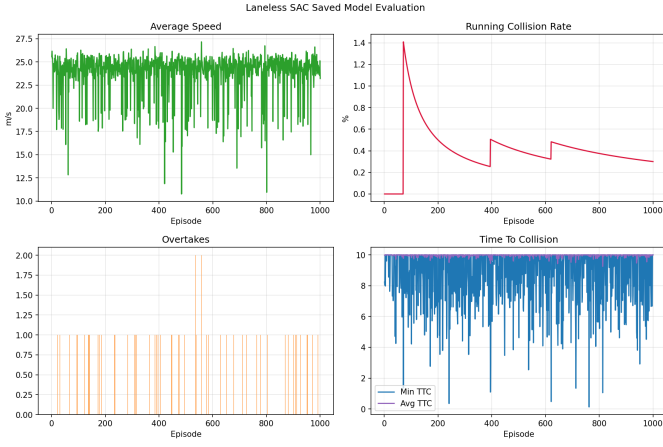
The laneless SAC agent was evaluated over 1000 deterministic episodes. The model achieved an average speed of 23.86 m/s, a collision rate of 0.30%, an average TTC of 9.97 s, and a minimum TTC of 8.74 s. These results indicate that the learned policy is able to drive safely in the laneless environment while maintaining highway-level speed.

The running collision rate remains close to zero throughout evaluation, showing that the policy is stable under the tested traffic distribution. The speed curve remains mostly between 23 and 25 m/s, meaning the agent does not solve the task by stopping or driving overly slowly.

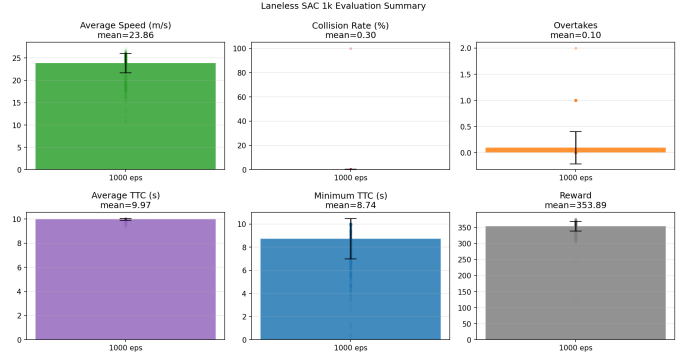
However, the policy remains conservative. The average number of overtakes is only 0.10 per episode, which suggests that the agent prioritizes safe cruising over active maneuvering. Therefore, the laneless SAC result should not be interpreted as a complete solution to aggressive laneless driving, but rather as evidence that the proposed laneless environment and continuous target-control formulation can produce a stable learned policy with low collision rate.

The SAC policy panels in Fig. 6 provide additional insight into the learned behavior. This confirms that the policy operates in the intended laneless action space: instead of choosing discrete lane changes, it selects continuous target-speed and lateral-position updates.

Overall, the laneless experiment validates the main formulation shift: lane selection is replaced by continuous corridor selection, and SAC can learn a stable policy in this action space. The remaining limitation is behavioral richness, so future work should focus on harder laneless traffic scenarios, stronger overtaking incentives, curriculum learning, and explicit lateral interaction metrics to encourage safe but more assertive corridor selection.



(a) Saved-model evaluation summary for the laneless SAC agent over 1000 episodes.



(b) Episode-level speed, collision rate, overtakes, and TTC trends for the laneless SAC agent.

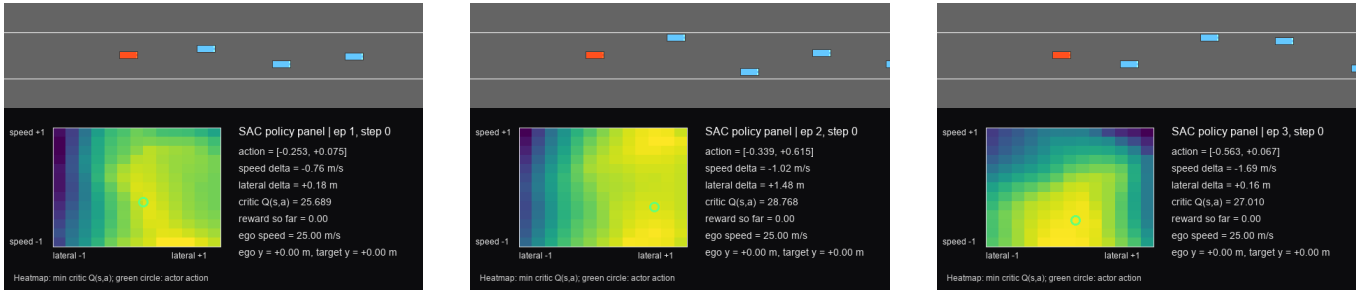


Fig. 6: Laneless SAC saved-model evaluation and policy panels. The green circle represents the actor-selected action over the critic value landscape.

VI. CONCLUSION

In this paper, we investigated reinforcement learning for decision-making in unstructured highway environments, focusing on congested traffic and laneless driving. These two scenarios show how reducing environmental structure naturally changes the problem formulation and forces new design choices in observation, reward, action space, and traffic modeling.

In congested traffic, we showed that a standard lane-based DQN formulation is not sufficient under dense interaction. Collision diagnostics revealed that the baseline policy failed mainly because it lacked early safety information and lateral risk awareness. TTC-based reward shaping improved longitudinal safety, attention improved dense-traffic representation when paired with safety shaping, and lane-context safety observations helped address the remaining side-collision failure mode. These results also suggest that hierarchical reinforcement learning may be an important future direction, since congested driving requires better separation between high-level maneuver choice and lower-level throttle or steering control.

In the laneless environment, we reformulated the highway problem by replacing discrete lane selection with continuous corridor selection. This required modifying the traffic model itself: lane-based IDM/MOBIL reasoning was generalized into corridor-based leader detection and continuous lateral target

selection. Using this formulation, SAC was applied to learn target-speed and target-lateral-position control, demonstrating that an RL agent can learn stable navigation behavior in a highway environment without explicit lane access.

Overall, unstructured environments act as an important stress test for current autonomous-driving RL methods. They expose where standard lane-based formulations break, but they also suggest how new formulations can extend existing methods. Future work should focus on integrating these unstructured-driving skills into general driving architectures, so that agents can move between normal structured driving, congestion, and laneless scenarios without requiring completely separate policies for each case.

REFERENCES

- [1] N. Wang, X. Li, K. Zhang, J. Wang, and D. Xie, "A survey on path planning for autonomous ground vehicles in unstructured environments," *Machines*, vol. 12, no. 1, p. 31, 2024.
- [2] B. Zhang, W. Zhan, L. Sun, J. Hu, and C. Tomlin, "A survey on motion planning for self-driving vehicles in scenarios with intersections," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 2, pp. 658–676, 2021.
- [3] C. Hubmann, M. Becker, D. Althoff, D. Lenz, and C. Stiller, "Decision making for autonomous driving considering interaction and uncertain prediction of surrounding vehicles," in *2017 IEEE Intelligent Vehicles Symposium (IV)*, 2017, pp. 1671–1678.
- [4] L. B. de la Cruz and H. Komurcugil, "Model predictive control for autonomous vehicle motion control: A review," *IEEE Transactions on Automation Science and Engineering*, 2023.

- [5] Y. Qian, X. Li, J. Wang, D. Cao, H. E. Tseng, and F. Borrelli, "Co-operative trajectory optimization for connected and automated vehicles at intersections using model predictive control," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 2, pp. 973–983, 2020.
- [6] Z. Wang, P. Wei, J. Zheng, H. Yin, and N. Xu, "Game-theoretic modeling of merging behaviors at highway on-ramps using model predictive control," *Transportation Research Part C: Emerging Technologies*, vol. 124, p. 102953, 2021.
- [7] T. Liu, B. Huang, Z. Deng, H. Wang, X. Tang, X. Wang, and D. Cao, "Heuristics-oriented overtaking decision making for autonomous vehicles using reinforcement learning," *IET Electrical Systems in Transportation*, 2020.
- [8] N. Sakib, "Highway lane change under uncertainty with deep reinforcement learning based motion planner," 2020, thesis or technical report.
- [9] A. Alizadeh, M. Moghadam, Y. Bicer, N. Ure, U. Yavas, and C. Kurtulus, "Automated lane change decision making using deep reinforcement learning in dynamic and uncertain highway environment," in *2019 IEEE Intelligent Transportation Systems Conference (ITSC)*, 2019, pp. 1399–1404.
- [10] F. Ye, X. Cheng, P. Wang, and C. Chan, "Automated lane change strategy using proximal policy optimization-based deep reinforcement learning," 2020.
- [11] Z. Cao, D. Yang, S. Xu, H. Peng, B. Li, S. Feng, and D. Zhao, "Highway exiting planner for automated vehicles using reinforcement learning," *IEEE Transactions on Intelligent Transportation Systems*, 2020.
- [12] J. Duan, S. E. Li, Y. Guan, Q. Sun, and B. Cheng, "Hierarchical reinforcement learning for self-driving decision-making without reliance on labelled driving data," *IET Intelligent Transport Systems*, vol. 14, no. 5, pp. 297–305, 2020.
- [13] C. Hoel, K. Driggs-Campbell, K. Wolff, L. Laine, and M. Kochenderfer, "Combining planning and deep reinforcement learning in tactical decision making for autonomous driving," *IEEE Transactions on Intelligent Vehicles*, vol. 5, no. 2, pp. 294–305, 2019.
- [14] Y. Wang, Z. Zhang, W. Zhan, J. Hu, C. J. Tomlin, and M. Tomizuka, "Safe reinforcement learning with model predictive control for autonomous lane merging," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 2843–2850, 2020.
- [15] X. Li, Y. Qian, J. Wang, D. Cao, H. E. Tseng, and F. Borrelli, "Deep reinforcement learning for coordinated ramp merging with model predictive control," in *2020 American Control Conference (ACC)*, 2020, pp. 485–490.
- [16] R. Bautista-Montesano, R. Galluzzi, K. Ruan, Y. Fu, and X. Di, "Autonomous navigation at unsignalized intersections: A coupled reinforcement learning and model predictive control approach," *Transportation Research Part C: Emerging Technologies*, vol. 139, p. 103662, 2022.
- [17] A. Liniger, A. Domahidi, and M. Morari, "Safe reinforcement learning using robust mpc for motion planning of autonomous vehicles," in *2019 IEEE 58th Conference on Decision and Control (CDC)*, 2019, pp. 8226–8232.
- [18] T. Zhao, Y. Sun, H. Chen, and N. Li, "Adaptive model predictive control for autonomous vehicle driving in different environments," in *2017 IEEE Intelligent Vehicles Symposium (IV)*, 2017, pp. 1421–1426.
- [19] B. Paden, M. Čáp, S. Z. Yong, D. Yershov, and E. Frazzoli, "A survey of motion planning and control techniques for self-driving urban vehicles," *IEEE Transactions on Intelligent Vehicles*, vol. 1, no. 1, pp. 33–55, 2016.
- [20] T. Liu, H. Wang, B. Lu, J. Li, and D. Cao, "Decision-making for autonomous vehicles on highway: Deep reinforcement learning with continuous action horizon," 2020.
- [21] E. Leurent, "An environment for autonomous driving decision-making," GitHub repository, 2018. [Online]. Available: <https://github.com/eleurent/highway-env>
- [22] B. R. Kiran, I. Sobh, V. Talpaert, P. Mannion, A. Sallab, S. Yogamani, and P. Pérez, "Deep reinforcement learning for autonomous driving: A survey," *IEEE Transactions on Intelligent Transportation Systems*, 2021.
- [23] S. Shalev-Shwartz, S. Shammah, and A. Shashua, "Safe, multi-agent, reinforcement learning for autonomous driving," in *5th ICML Workshop on Machine Learning for Autonomous Vehicles*, 2016.
- [24] J. Bae, K. Lee, S. Lee, S. C. Lee, and S.-H. Son, "Safe reinforcement learning for autonomous driving at unsignalized intersections with safety-critical control," *IEEE Robotics and Automation Letters*, vol. 8, no. 5, pp. 2556–2563, 2023.
- [25] R. Zhang, X. Xue, C.-Z. Lin, J. Sun, and P. Stone, "Learning feasible and adaptive model predictive control for interactive autonomous driving with safety constraints," *IEEE Robotics and Automation Letters*, vol. 8, no. 8, pp. 4793–4800, 2023.
- [26] F. Dang, D. Chen, J. Chen, and Z. Li, "Event-triggered model predictive control with deep reinforcement learning for autonomous driving," *IEEE Transactions on Intelligent Vehicles*, 2023.
- [27] B. Brito, A. Agarwal, and J. Alonso-Mora, "Learning interaction-aware guidance for trajectory optimization in dense traffic scenarios," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 10, pp. 18 808–18 821, 2022.
- [28] W. Song, G. Xiong, and H. Chen, "Intention-aware autonomous driving decision-making in an uncontrolled intersection," *Mathematical Problems in Engineering*, 2016.
- [29] I. Mirchevska, M.-L. Ta, V. D. Hoang, and C. Stiller, "Reinforcement learning for autonomous driving at intersections with intention-aware q-networks," in *2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC)*, 2022, pp. 3790–3797.
- [30] E. Leurent and J. Mercat, "Social attention for autonomous decision-making in dense traffic," 2019.
- [31] J. Nie, J. Zhang, E. Ding, X. Wan, X. Chen, and B. Ran, "Decentralized cooperative lane-changing decision-making for connected autonomous vehicles," *IEEE Access*, vol. 4, pp. 9413–9420, 2016.
- [32] L. Li, K. Ota, and M. Dong, "Humanlike driving: Empirical decision-making system for autonomous vehicles," *IEEE Transactions on Vehicular Technology*, vol. 67, no. 8, pp. 6814–6823, 2018.
- [33] M. Treiber, A. Hennecke, and D. Helbing, "Congested traffic states in empirical observations and microscopic simulations," *Physical Review E*, vol. 62, pp. 1805–1824, 2000.
- [34] A. Kesting, M. Treiber, and D. Helbing, "General lane-changing model mobil for car-following models," *Transportation Research Record*, vol. 1999, no. 1, pp. 86–94, 2007.
- [35] M. Zhou, J. Luo, J. Vilella, Y. Yang, D. Rusu, J. Miao, W. Zhang, M. Alban, I. Fadakar, Z. Chen, A. C. Huang, Y. Wen, K. Hassanzadeh, D. Graves, D. Chen, Z. Zhu, N. Nguyen, M. Elsayed, K. Shao, S. Ahilan, B. Zhang, J. Wu, Z. Fu, K. Rezaee, P. Yadmellat, M. Rohani, N. P. Nieves, Y. Ni, S. Banijamali, A. C. Rivers, Z. Tian, D. Palenicek, H. B. Ammar, H. Zhang, W. Liu, J. Hao, and J. Wang, "Smarts: Scalable multi-agent reinforcement learning training school for autonomous driving," in *Proceedings of the 4th Conference on Robot Learning*, 2020.
- [36] J. Bernhard, K. Esterle, P. Hart, and T. Kessler, "Bark: Open behavior benchmarking in multi-agent environments," in *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020.
- [37] S. Rahmani, Z. Xu, S. C. Calvert, and B. van Arem, "A bi-level real-time microsimulation framework for modeling two-dimensional vehicular maneuvers at intersections," in *2023 IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)*, 2023, pp. 4221–4226.
- [38] M. L. Puterman, *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. John Wiley & Sons, 2014.
- [39] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017.
- [40] J. Schulman, S. Levine, P. Abbeel, M. Jordan, and P. Moritz, "Trust region policy optimization," in *International Conference on Machine Learning*, 2015, pp. 1889–1897.
- [41] R. Rajamani, *Vehicle Dynamics and Control*. Springer, 2011.
- [42] J. Kong, M. Pfeiffer, G. Schildbach, and F. Borrelli, "Kinematic and dynamic vehicle models for autonomous driving control design," in *2015 IEEE Intelligent Vehicles Symposium (IV)*, 2015, pp. 1094–1099.